Tangents and normals

1 Find the gradient of the normal to the curve $y = x^2 + \cos(\pi x)$, at the point where $x = 1$.

2 For the following functions, find the equation of the tangent at the given point:

a $y = -5x \sin x$ at $(\pi, 0)$.

b $y = \cos 6x$ at the point where $x = \frac{\pi}{18}$.

3 Consider the function $f(x) = \cos 2x - \sqrt{3} \sin 2x$.

a Find the first positive value of x at which the curve intersects the x-axis.

b Find the gradient of the tangent at this point.

c Hence determine the positive angle, θ, that the tangent makes with the positive x-axis at this point. Round your answer to the nearest degree.

4 Consider the function $f(x) = -4 \cos 3x$.
Find θ, the angle that the tangent at $x = \frac{\pi}{12}$ makes with the positive x-axis. Round your answer to the nearest degree.

5 For each of the following functions, find:

i The derivative function.

ii The gradient of the normal at the point $\left(\frac{\pi}{4}, 1\right)$.

iii The equation of the normal at the point $\left(\frac{\pi}{4}, 1\right)$.

a $y = \sin 2x$ **b** $y = \sqrt{2} \sin x$ **c** $y = \sqrt{2} \cos x$

6 For the function $y = \frac{\sin x}{x}$, find:

a The derivative function.

b The gradient of the normal at the point $\left(\frac{\pi}{2}, \frac{2}{\pi}\right)$.

c The equation of the normal at the point $\left(\frac{\pi}{2}, \frac{2}{\pi}\right)$.

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Trigonometric graphs

7 Consider the function $y = \dfrac{\sin x}{1 + \cos x}$.

a Find $\dfrac{dy}{dx}$.

b Determine the number of turning points function y has.

8 Find the x-coordinate of each point of inflection on the curve $y = 3\cos\left(2x + \dfrac{\pi}{4}\right)$ for the interval $0 \leq x \leq 2\pi$.

9 Consider the function $f(x) = -4\sin\left(x + \dfrac{\pi}{6}\right)$ on the interval $0 \leq x \leq 2\pi$.

a Find the coordinates of any turning points.

b Find an expression for $f''(x)$.

c Hence classify the turning points.

10 For each of the following functions on the interval $0 \leq x \leq 2\pi$:

i Find the axes intercepts.

ii Find an expression for $f'(x)$.

iii Determine the coordinates of any stationary points.

iv Find an expression for $f''(x)$.

v Hence classify the stationary points.

vi Sketch the graph of $f(x)$.

a $f(x) = 2\sin x$ **b** $f(x) = \cos 3x$ **c** $f(x) = x\cos x$ **d** $f(x) = e^{-x}\cos x$

e $f(x) = \sin^2(x)$ **f** $f(x) = \cos^3(x)$ **g** $f(x) = x + \sin x$

Applications

11 Determine the possible values of k, given that $y = \cos kx$ satisfies the following equation for all values of x:

$$y'' + 25y = 0$$

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12 The displacement of an object is a measure of how far it is from a fixed origin point at time, t seconds. Displacement to the left of the origin point is negative and displacement to the right of the origin is positive. At time t seconds, velocity is given by $v(t) = x'(t)$, and acceleration is given by $a(t) = x''(t)$.
If the displacement of an object x about a fixed origin is given by:

$$x(t) = -5\cos 2t$$

a Describe the initial position of the object, from the fixed origin point.

b Write an expression for $v(t)$.

c Show that $a(t) = -4x(t)$.

d Find the acceleration of the object, when $t = \dfrac{\pi}{2}$.

13 A particle moves in a straight line and its displacement, x cm, from a fixed origin point after t seconds is given by the function:

$$x(t) = \sin t - \sin t \cos t - 5t$$

a State the initial displacement of the particle.

b The velocity is the rate at which displacement changes over time. Find an equation for $v(t)$, in terms of $\cos t$.

c Hence find the initial velocity, $v(0)$.

14 The arm of a pendulum swings between its two extreme points A to the left and B to the right. Its horizontal displacement x cm from the centre of the swing, at time t seconds after it starts swinging, is given by:

$$x(t) = 16\sin 3\pi t$$

a Find the intial position of the pendulum.

b Find the maximum distance of the pendulum from the central position of its swing.

c Find an expression for the velocity function, $v(t) = x'(t)$.

d Find the first two times at which the pendulum comes to rest, $v(t) = 0$.

e Find the displacement of the pendulum when it first comes to rest.

f Find the displacement of the pendulum when it comes to rest for the second time.

g Hence determine the distance between the two points A and B.

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15 The arm of a pendulum swings between its two extreme points A to the left and B to the right. Its horizontal displacement x cm from the centre of the swing at time t seconds after it starts swinging is given by:

$$x(t) = 19 \sin 4\pi t$$

a Find an expression for the velocity function, $v(t) = x'(t)$.

b Find the maximum velocity of the pendulum.

c Find the first two times, after it starts swinging, at which the pendulum reaches its maximum velocity, $v'(t) = 0$ and $x'(t) < 0$.

d Describe the position of the pendulum when it reaches its maximum velocity.

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